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Research article

Analysis of spatial differences in global regional human development index under planetary pressure and decomposition study of driving factors

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ABSTRACT

To urge nations worldwide to implement robust measures for enhancing human development and mitigating the pressures exerted on the planet by human activities in pursuit of sustainable development, this study encompasses 154 countries globally, using the seven major regions as focal points. Leveraging the Planetary Pressure Adjusted Human Development Index (PHDI) proposed by the United Nations Development Programme on December 15, 2020, as a metric for human development levels, this paper aims to standardize and internationally compare human development data from 1990 to 2021. Subsequently, employing the Theil index, the study assesses the global human development status across the seven regions to analyze spatial disparities in PHDI. Lastly, a comprehensive Generalized Diese Index Method (GDIM) is constructed to accurately reflect absolute and latent factors, dissecting the driving forces impacting global PHDI. The study explores critical pathways for high-quality human development within the harmonious coexistence between humanity and nature. It validates the robustness of GDIM results through a stepwise regression. Research findings indicate varying levels of PHDI development across regions, with a distinct spatial hierarchy evolving: higher human development levels in Europe and Eastern Europe, favorable levels in North and South America, similar levels in Oceania and Asia, and significant improvement potential in Africa. As globalization progresses, overall differences in PHDI gradually decrease; however, disparities persist between and within regions. Economic, technological, and per capita welfare effects consistently positively drive PHDI. In contrast, environmental pressure effects, social effects, per capita value-added effects, and output carbon intensity effects consistently exert hostile driving forces. Population size effects on PHDI show a fluctuating trend. Moreover, in terms of cumulative contribution values, the top three contributors to driving forces are economic, technological, and per capita welfare effects.

1. Introduction

In the context of humankind's and the planet's entrance into a new geological era, nations have accomplished noteworthy advancements in enhancing the well-being of their populations. Nevertheless, the elevation of human development levels has concurrently engendered challenges, including resource depletion, environmental contamination, global warming, and extreme weather events, which have precipitated substantial ecological, economic, social, and resource strains on our planet. In 2019, the outbreak of the COVID-19 pandemic posed a severe crisis to our world, serving as a stark warning to humanity. It underscored the imperative for humanity to cease its voracious exploitation of nature, as such actions could precipitate even greater calamities for the planet. Furthermore, pollution, armed conflicts, and energy crises are pervasive global realities (Simionescu et al., 2023), presenting formidable challenges to human development. It's exacerbating the planetary pressures.

In 2020, Achim Steiner, the Administrator of the United Nations Development Programme, emphasized that humanity is exerting unprecedented pressure on the planet. He underscored the imperative for society to consider factors such as environmental pressures as one of the dimensions for measuring development and progress. Presently, the prevailing approach to gauge and compare the relative levels of human development across nations predominantly relies on the Human Development Index (HDI) (Noorbakhsh, 1998; Decancq et al., 2009; Li and Zhuang, 2008). This index solely encompasses the dimensions of health, education, and income without yet incorporating planet pressures into the assessment of human development, potentially resulting

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in limitations within the measurement data. Furthermore, a significant improvement in human development does not necessarily imply equitable human development among global regions and nations. Over the long term, such imbalances in human development will likely decelerate the overall progress of international human development, resulting in stagnation or regression in certain countries and regions (United Nations Development Programme, 2014).

Hence, the research objective of this study embarks on an investigation from the perspective of accurately gauging and assessing the global human development process, conducting an in-depth analysis of regional disparities in human development, and unveiling the drivers behind human development levels. The significance of this research is that it not only facilitates the mobilization of governments worldwide to counteract the prevailing antagonistic dynamics between humans and nature, thereby advancing the next frontier of human development, but also promotes the swift modification of societal norms, values, and governmental incentive measures. Such endeavors aim to alleviate the pressures exerted on the planet by humanity, ensuring high-quality global human development.

This paper is structured as follows: the second part reviews the research literature on human development level assessment, regional differences, and driving mechanisms. The third part introduces model theory and methods, focusing on the theory related to measuring the Planetary Pressures-Adjusted Human Development Index (PHDI), the Thiel coefficient, and the Generalized Divisia Index Method (GDIM). The fourth part provides empirical analyses of PHDI for regional differences analysis and driver decomposition studies. The sixth part discusses the relevant literature. The seventh part puts forward policy recommendations based on the conclusion and the forcing mechanism and points out the limitations of the paper and future research directions.

2. Literature review

Assessing human development levels across various countries has become a central focus of scholarly research. While domestic and international scholars have widely employed the HDI to measure and compare the relative levels of human development in different countries, scholars have gradually come to recognize the influence of environmental factors on human development (Kummu et al., 2018). Some scholars argue that the HDI fails to consider the impact of ecological environments, resource consumption, and other factors on human development, highlighting that its indicator framework remains incomplete (Li, 2007; Bravo, 2014). In this context, some scholars advocate for including ecological environmental indicators within the HDI. Lasso and Urruia (2001) argue that the influence of the environment on human development should not be overlooked and propose the recalibration of HDI through the incorporation of per capita carbon emissions. Bravo (2014) further asserts that the HDI should account for the environmental dimension by introducing ecological environmental factors as a fourth dimension with equal weight within the HDI framework.

On the other hand, another group of scholars advocates for considering resource consumption indicators within the HDI. Wang and Jiang (2009) suggest revising the HDI from the resource consumption perspective by incorporating indicators such as energy consumption per unit GDP, comprehensive utilization of industrial solid waste, and compliance rate of industrial wastewater discharge standards. Furthermore, some scholars argue for considering both resource consumption and environmental indicators simultaneously. Tian et al. (2007) introduced ecological pollution and energy consumption indicators primarily from the perspective of pollution emissions to modify the HDI. Li and Wang (2020) underscore the importance of constructing an ecologically sensitive regional HDI for China by incorporating two dimensions: resource consumption and pollution emissions. In this regard, HDI can somewhat reflect the current situation. However, it is not a perfect indicator and cannot accurately measure human development. Using it directly for international comparisons may have certain drawbacks. At the same time, as humanity and the planet enter a new geological epoch, a series of issues, such as resource depletion and environmental pollution, urgently need to be addressed. The pressure on the planet cannot be ignored, and countries urgently need to take responsibility for the dangerous pressure humans exert on the planet and make changes (United Nations Development Programme, 2020). Despite scholars having proposed amendments to the HDI, further exploration is needed to assess the rationality of the correction methods and indicator selection. Therefore, in measuring human development levels, it is necessary to incorporate planet pressure factors such as the ecological environment and resource consumption into the HDI to urge countries to take robust measures to alleviate the pressure on the planet.

In human development studies, the interconnectedness of human development levels across various regions plays a pivotal role in shaping the overall trajectory of human progress. Researchers in the field of human development predominantly utilize HDI as a primary metric for investigation, with a particular focus on examining disparities at the urban-rural, inter-provincial, or national levels. For instance, Song and Ma (2004) conducted pioneering research using the HDI to measure China's urban-rural disparities in human well-being. Similarly, Bian and Wang (2021) evaluated the Human Green Development Index for Liaoning province from 2009 to 2019. Meanwhile, Hu et al. (2018) analyzed variance decomposition to explore regional disparities in human development levels and the underlying factors contributing to regional disparities.

Furthermore, Marchante et al. (2006) conducted an in-depth analysis of the evolution of human development in Spain from 1980 to 2001. In a more recent study, Zhu et al. (2021) employed Exploratory Spatial Data Analysis (ESDA) methods to examine the spatiotemporal characteristics of HDI across 31 provinces from 1990 to 2017. Their work delved into the dynamic mechanisms and determining factors influencing the HDI. These studies collectively underscore the significance of assessing human development disparities at various geographical scales and utilizing HDI as a valuable tool to comprehend the multifaceted aspects of human well-being and progress. In general, existing literature primarily focuses on applying HDI in analyzing regional disparities, emphasizing the provision of development measures based on differences in index values. Few studies have incorporated HDI and regional disparity characteristics to conduct regional investigations into the modification or reconstruction of the HDI. Furthermore, domestic research has examined chiefly the imbalance in human development at the provincial level, with limited studies considering the evolution of the imbalance in human development patterns at a sub-regional level on a global scale (Wang and Zhai, 2018). Therefore, from a global and regional perspective, using a modified HDI to explore the mechanisms underlying the evolution of human development disparities becomes exceptionally crucial.

In the research on drivers of human development levels, Shan and Cheng (2017), Li et al. (2018), Hu et al. (2018), and Wang and Jiang (2020) have each employed spatial econometric analysis, the Tapio decoupling model, variance decomposition, and the Logarithmic Mean Divisia Index (LMDI) method, respectively, to analyze the factors driving human development levels, yielding divergent conclusions. However, significant gaps remain in the comprehensive study of the drivers of human development levels, characterized by the following deficiencies: Firstly, the selection of driver indicators is not yet comprehensive. Chen and Chen (2015) systematically analyzed the impact of public fiscal policies and income inequality on China's human development levels. Ren et al. (2020) employed panel models to explore the influence of factors such as education, social security initiatives, and disposable income of urban and rural residents on human development levels. Redmond and Nasir (2020) pointed out that the abundance of natural resources and the degree of trade openness have negative and positive effects on human development, respectively. Opoku et al. (2022) found that reducing the ecological footprint, total carbon dioxide, greenhouse gas emissions, and exposure to air pollution can improve human development levels, thereby contributing to greater environmental sustainability. Secondly, there are shortcomings in the driver factor decomposition methods. Currently, the comprehensive study of factors driving human development levels remains incomplete. Wang and Jiang (2020), Wang and Yu (2020) have successfully utilized the extended Kaya identity and LMDI factor decomposition techniques to identify the driving effects of China's human development changes, emphasizing that different factors have varying degrees of impact on improving human development levels. However, these decomposition methods exhibit certain limitations. Vaninsky (2014) has pointed out that existing index decomposition methods predominantly rely on the Kaya identity, decomposing the target variable into a product of multiple factors. As driving factors interdepend, the decomposition results depend on selecting influencing factors, potentially yielding conclusions that diverge from reality. Furthermore, this method can only examine the influence of individual absolute factors. It cannot reveal the impact of changes in other absolute factors and latent variables on variations in human development. Therefore, for a comprehensive analysis that accurately reflects the impact of driving factors on human development levels and to achieve regional development balance sooner, it is essential to combine developmental mechanisms. It involves improving the selection of driver factor indicators and choosing driver factor decomposition methods that can adequately consider both absolute and latent factors.

In general, the research outcomes about the assessment of human development levels by scholars both domestically and internationally have been significant. However, this paper posits that there are still deficiencies in the following three aspects: Firstly, the evaluation system for human development levels is imperfect, and there is a lack of uniformity in the indicators and methods used to amend HDI. Currently, HDI remains the most frequently employed measure in research, yet it fails to adequately address the trajectory of human development, often neglecting the influence of planetary pressure factors. Although a minority of scholars have adjusted the HDI, disputes still exist regarding selecting indicators and measurement methods. Secondly, there are limitations in the scope of research areas. Existing studies predominantly focus on analyzing imbalanced urban-rural, interprovincial, or within a single country's human development disparities, with a shortage of research outcomes concerning regional disparities in global human development levels. Thirdly, limitations exist in studying driving factors, including their indicators and methodologies. On the one hand, the selection of research indicators is not sufficiently comprehensive. On the other hand, many decomposition methods for driving factors are grounded in the Kaya identity, with their decomposition outcomes also contingent upon the choice of influencing factors. It can result in decomposition conclusions that may diverge from reality due to potential contradictions arising from different factor decomposition forms.

Therefore, this paper aims to improve the following aspects: Firstly, enhancing the construction framework of HDI. The measurement method employed will utilize the recently proposed PHDI by the United Nations Development Programme. This approach comprehensively accounts for planetary pressures, evaluating a nation's achievements in three fundamental dimensions of human development: health, knowledge, and resource acquisition. The goal is to advance human development initiatives with reduced emissions and resource utilization. Secondly, it investigates global human development disparities across 154 countries using the Theil index. This paper employs Theil index analysis to assess spatial disparities in human development levels among the 154 countries. These nations are further categorized into seven major regions to explore the evolving patterns and mechanisms of balance across these global regions. Thirdly, to address the prevailing human-nature antagonism and promote the next frontier of human development geared towards high quality of life, this study utilizes the GDIM to decompose PHDI, considering multiple influencing factors. The research examines the effects of the decomposed driving factors on PHDI.

3. Research methodology

3.1. PHDI measurement methodology

Human activities have exerted significant pressure on the planet's environment, including climate change, resource depletion, and ecological degradation. To better understand the challenges humanity faces in alleviating global pressures, unveil shifts in the global development landscape, scientifically assess human development and achievements across nations, study human development levels, and formulate more sustainable development strategies. In December 2020, the United Nations Development Programme introduced the PHDI as a derivative index of the HDI. The PHDI accurately measures a nation's achievements in three fundamental dimensions of human development—health, knowledge, and resource acquisition—within the context of planetary pressures. It aims to achieve a high level of human development while minimizing emissions and resource usage. Compared to the HDI and its adjusted indices, the PHDI holds distinct advantages in addressing measurement issues arising from excluding planetary pressure factors, such as resource consumption and environmental pollution. Furthermore, it reflects three human development choices under planetary pressures: a healthy life, knowledge acquisition, and resource attainment for a dignified existence. This index defines human well-being and planetary pressure mitigation as central to human development, effectively promoting global development and carrying a certain level of authority. Moreover, the PHDI encompasses cross-cutting issues in critical domains like sustainable development, environmental conservation, social equity, and human welfare.

Theoretically, the PHDI should be equal to the HDI in an ideal scenario without planetary pressures. However, the PHDI often falls below the HDI as planetary pressures increase. Furthermore, calculating the PHDI measures human development levels and clarifies humanity's challenges in mitigating planetary pressures. It effectively encourages human action to reduce planetary pressures, thereby promoting improvements in human development levels. In summary, this index possesses numerous advantages. In this study, we intend to measure the PHDI for countries globally from 1990 to 2021 using the current calculation method to enhance the study of human development levels. The specific process of PHDI formation is depicted in Fig. 1.

First, we must calculate the HDI, a composite index comprising health, education, and income.

It is calculated using the formula.

$$HDI = (I_{health} \times I_{education} \times I_{income})^{\frac{1}{3}} \tag{1}$$

Among them, the health dimension I_{health}, measured by life expectancy at birth, the education dimension $I_{education}$, where the use of knowledge is expressed through the degree of educational attainment, which is a composite indicator of adult literacy and gross enrollment rate, and the income dimension, calculated using real Gross National Income (GNI) per capita adjusted by purchasing power parity (PPP).

Second, the PHDI adjusts the level of human development based on two important indicators: carbon dioxide emissions per capita and material footprint per capita. The following formulas normalize carbon emissions per capita and material footprint per capita.

$$MF = \frac{max\ MF_j - valueMF}{maxMF_j - min\ MF_j} \tag{2}$$

$$Carbon = \frac{max\ CO2_j - valueCO2}{max\ CO2_j - min\ CO2_j} \tag{3}$$

MF denotes the material footprint coefficient per capita. $max\ MF_j$ and $min\ MF_j$ denote the maximum and minimum values of the historical observed per capita material footprint for all countries since 1990. $valueMF$ indicates the material footprint per capita. $Carbon$ denotes the per capita carbon emission factor. $max\ CO2_j$ and $minCO2_j$ denote the

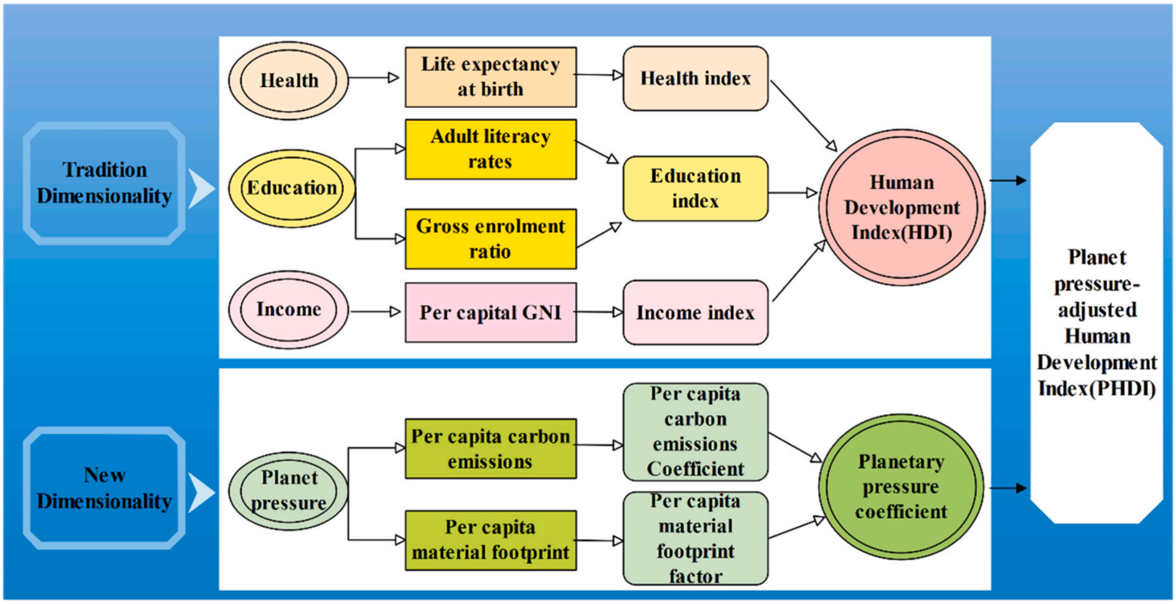

Fig. 1. PHDI construction.

maximum and minimum values of historical observed per capita carbon emissions for all countries since 1990. $valueCO2$ denotes carbon emissions per capita.

The United Nations Development Programme defines 0 as the minimum value for per capita material footprint and per capita carbon emissions. The maximum value for per capita material footprint corresponds to the highest recorded value of 107.42 in Kuwait in 1996. Likewise, the maximum value for per capita carbon dioxide emissions corresponds to Qatar's highest recorded value of 68.72 metric tons in 1997 (Biggeri and Mauro, 2018). It is worth noting that the ranking of countries is highly sensitive to the choice of these maximum values.

Therefore, the normalized equation in this paper is:

$$MF = \frac{107.42 - valueMF}{107.42} \tag{4}$$

$$Carbon = \frac{68.72 - valueCO2}{68.72} \tag{5}$$

Next, the adjustment factor for planetary pressures is calculated as A. The formula is

$$A = \frac{Carbon + MF}{2} \tag{6}$$

Overall, regarding carbon emissions per capita and material footprint per capita, the higher the observed value and the closer to the maximum value, the higher the pressure on the planet, which also means that the smaller the value of the carbon emissions per capita factor and material footprint per capita factor, the greater the adjustment to the HDI. Finally, the PHDI calculation formula can be obtained as follows.

$$\begin{aligned} PHDI = HDI \times A &= (I_{\text{health}} \times A \times I_{\text{education}} \times A \times I_{\text{income}} \times A)^{\frac{1}{3}} \\ &= (PI_{\text{health}} \times PI_{\text{education}} \times PI_{\text{income}})^{\frac{1}{3}} \end{aligned} \tag{7}$$

where PI_{health}, $PI_{\text{education}}$ and PI_{income} represent the planetary pressures-adjusted health, education, and income indexes, respectively.

3.2. Theil index and its spatial decomposition

The Theil index, introduced by Theil in 1967, is commonly employed in multidimensional studies to measure regional disparities based on a single indicator. Its advantages lie in its ability to intuitively assess the degree of differentiation among different social groups and reflect differences within and between regions, thus explicitly identifying their contributions to the total disparity. It, in turn, assists policymakers in identifying and addressing issues of inequity. Higher values of the Theil index signify greater inequality, while lower values indicate a more equitable distribution. The Theil index possesses intuitiveness, comparability, decomposability, and mathematical properties. These attributes provide crucial references and guidance for policy formulation, research, and evaluation, contributing to the pursuit of more equitable and sustainable development.

$$T_{theil} = \sum \frac{y_m}{Y} \times lg \frac{(y_m / Y)}{(x_m / X)} \tag{8}$$

T_{theil} denotes the Theil index, x_m and y_m denote the total population and PHDI of the ith country in the region of m. X and Y denote the total global population and total PHDI, respectively. In general, T_{theil} takes values between [0,1], with larger coefficient values indicating more significant regional differences in human development levels.

In addition, the overall regional imbalance can be decomposed into an intra-regional and inter-regional variation based on the Theil index.

$$T_{theil} = T_{inter} + T = T_{inter} + \sum \frac{Y_i}{Y} \times T_i \tag{9}$$

$$T_{inter} = \sum \frac{Y_i}{Y} \times lg \frac{(Y_i / Y)}{(X_i / X)} \tag{10}$$

$$T_i = \sum \frac{y_m}{Y} \times lg \frac{(y_m / Y_i)}{(x_m / X_i)} \tag{11}$$

$$Y_i = \sum y_m \quad m \in i \; ; i = 1 \; , 2 \; , 3 \cdots 7 \tag{12}$$

$$X_i = \sum x_m \quad m \in i \; ; i = 1 \; , 2 \; , 3 \cdots 7 \tag{13}$$

where x_i and y_i denote the total population and total PHDI for i -*th* country, respectively. X_i and Y_i denote total population and total PHDI for i -*th* region, respectively; T_{inter} denotes inter-regional variation, T denotes intra-regional variation, and T_i denotes intra-regional variation for i -*th* region.

3.3. GDIM driver decomposition model

The GDIM, initially proposed by Vaninsky (2014), is constructed based on a modified Kaya identity version, incorporating absolute and relative factors in a multidimensional factor decomposition model. GDIM enables a more comprehensive and accurate quantification of the actual contributions of different factors to the evolution of the research variable, thereby revealing the driving forces behind the changes in the research variable. Prior scholars have found that index decomposition methods can, at most, examine a single absolute quantity factor and fail to reveal the impact of changes in other absolute quantity factors on the variation of the research variable. Moreover, implicit influencing factors are challenging to consider during decomposition. GDIM overcomes some significant shortcomings of existing index decomposition methods, making it an effective tool for more comprehensively and accurately quantifying the actual contributions of different factors to the evolution of the research object.

Specific advantages of GDIM include: First, GDIM not only decomposes the target variable into a product of multiple factors and handles the interdependencies among these factors. It means that GDIM's decomposition results are not influenced by factor selection and can consider the complex interactions among various factors, providing a more accurate revelation of the fundamental reasons for the variation in the target variable. Second, GDIM addresses the inadequacy of the existing index decomposition methods regarding the formal dependence among factors and distinguishes the decomposition results according to the relationships among all factors. This method expands the scope of analysis, avoids redundancy in calculations, and examines the influence of latent factors on the variation of the research variable. It exhibits a certain level of sophistication and aids in guiding policy formulation and environmental management.

Based on GDIM, the expressions for PHDI and its associated influences are as follows:

$$PHDI = (PHDI / GDP) \times GDP = (PHDI / CO_2) \times CO_2 = (PHDI / P) \times P \tag{14}$$

$$GDP / P = (PHDI / P) / (PHDI / GDP) \tag{15}$$

$$CO_2 / GDP = (PHDI / GDP) / (PHDI / CO_2) \tag{16}$$

To simplify the equation, let $Z = PHDI$, $X_1 = GDP$, $X_2 = PHDI/GDP$, $X_3 = CO_2$, $X_4 = PHDI/CO_2$, $X_5 = P$, $X_6 = PHDI/P$, $X_7 = GDP/P$ and $X_8 = CO_2/GDP$. Z denotes the PHDI, X_1 denotes the economic effects, the leading indicator of economic growth, in United States dollars. X_2 denotes the social effects, reflecting the contribution of economic growth to the level of development of the PHDI. X_3 denotes the environmental pressure effects in the ton. X_4 denotes the technological effects, reflecting the impact of technological progress on the level of development of the PHDI. X_5 denotes the population size effects on several people. X_6 denotes the per capita welfare effects, reflecting the contribution of population change to the level of PHDI development. X_7 denotes the per capita value-added effects in United States dollars. X_8 denotes the output carbon effects, reflecting the economic output resulting from the consumption of carbon-emitting resources within an economy. Further, Eqs. (14)–(16) are deformed as

$$Z = X_1X_2 = X_3X_4 = X_5X_6 \tag{17}$$

$$X_7 = X_6/X_2 \tag{18}$$

$$X_8 = X_2/X_4 \tag{19}$$

And from Eqs. (17)–(19) can be further transformed into Eqs. (20)–(24) .

$$Z = X_1X_2 \tag{20}$$

$$X_1X_2 - X_3X_4 = 0 \tag{21}$$

$$X_1X_2 - X_5X_6 = 0 \tag{22}$$

$$X_1 - X_5X_7 = 0 \tag{23}$$

$$X_3 - X_1X_8 = 0 \tag{24}$$

Let the contribution of the relevant factors to the change in PHDI be expressed as $\nabla Z = (X_2, X_1, 0, 0, 0, 0, 0, 0)^T$, and from Eqs. (20)–(24), a Jacobi matrix φ_X of relevant factors can be constructed.

$$\varphi_X = \begin{pmatrix} X_2 & X_1 & -X_4 & -X_3 & 0 & 0 & 0 & 0 \\ X_2 & X_1 & 0 & 0 & -X_6 & -X_5 & 0 & 0 \\ 1 & 0 & 0 & 0 & -X_7 & 0 & -X_5 & 0 \\ -X_8 & 0 & 1 & 0 & 0 & 0 & 0 & -X_1 \end{pmatrix}^T \tag{25}$$

According to the GDIM, the amount of change in the PHDI can be decomposed into the form of the sum of the contributions of the following factors:

$$\Delta Z[\varphi_X] = \int_t \nabla Z^T \left(I - \varphi_X \varphi_X^{-1} \right) dx \tag{26}$$

where t is the period, I is the unit matrix, and if the Jacobi matrix φ_X is linearly independent, the generalized inverse matrix $\varphi_X^{-1} = (\varphi_X^T \varphi_X)^{-1} \varphi_X^T$. The PHDI can be decomposed into ΔZ_{X1}, ΔZ_{X2}, ΔZ_{X3}, ΔZ_{X4}, ΔZ_{X5}, ΔZ_{X6}, ΔZ_{X7} and ΔZ_{X8}. The three absolute influences ΔZ_{X1}, ΔZ_{X3}, ΔZ_{X5} represent the effects of changes in the size of the economy, changes in environmental pressures, and changes in the size of the population on changes in the level of development of the PHDI, respectively, and the relative influences ΔZ_{X2}, ΔZ_{X4}, ΔZ_{X6}, ΔZ_{X7}, ΔZ_{X8} represent the effects of changes in social effects, changes in technological effects, changes in per capita welfare effects, changes in per capita value-added effects, and changes in output carbon emissions effects on changes in the level of development of the PHDI, respectively.

In this paper, considering the indicators' scientific, stability, and operability, we consider the actual situation of the available statistics, where some countries (or regions) may have missing data. Therefore, we first screen the collected data, followed by interpolation for countries with less missing data. After excluding countries with significant data gaps, the final study was conducted with 154 countries worldwide, with a panel data sample from 1990 to 2021. The HDI data is sourced from the United Nations Development Programme, the GDP from the International Monetary Fund, the number of countries from the World Human Development Report, the carbon emissions data from the Global Carbon Project, and the material footprint per capita data from the United Nations Environment Programme.

4. Empirical analysis

The global landscape exhibits heterogeneity in regional economic development, energy resource endowment, and geographical positioning, resulting in discernible disparities in human development indices across regions. Consequently, this study adopts a regional-centric approach to comprehensively scrutinize these regional differentials in PHDI and distill overarching regional development principles. Building upon the framework proposed by Wang and Zhai (2018), the world is delineated into seven distinct regions: Asia, Africa, North America, South America, Oceania, Europe, and Eastern Europe. Specifically, the Asian region encompasses 44 nations, including prominent examples such as China and Turkey, while the African region encompasses 45 nations, such as Egypt and Niger. The North American region comprises 14 countries, including the United States and Canada, and the Oceania region encompasses three countries: Australia and New Zealand. The South American region encompasses 10 countries, including Brazil and Peru, while the Eastern European region encompasses 15 countries, including Russia and Ukraine. Finally, the European region includes 23 countries other than Eastern Europe, including Denmark and Italy.

4.1. Basic overview of PHDI across regions

Regarding regional analysis, the PHDI exhibits varying levels of development across different regions (see Fig. 2). However, the overall trend shows a consistent pattern of gradual decline from highly developed areas to developing ones. The predominant range of PHDI values falls within the 0.500 to 0.800 range. This spatial distribution pattern reveals a two-tiered structure, with regions in developed countries generally displaying higher levels of human development. In comparison, regions in developing countries tend to exhibit lower human development levels. Furthermore, distinct spatial dynamics are observed in the evolution of PHDI levels across regions. Specifically, a pattern emerges where European and Eastern European regions exhibit relatively higher levels of human development. North and South American regions demonstrate comparatively favorable human development levels. And Oceania and Asian regions display relatively similar levels of human development. African regions present significant potential for improvement in human development levels.

In the PHDI analysis context, Europe and Eastern Europe have consistently maintained elevated levels. A comprehensive assessment of regional development conditions reveals that these two prototypical regions predominantly consist of developed countries. These nations exhibit well-established education and healthcare infrastructures, possess highly developed economies, and benefit from geographic proximity to one another and adjacent maritime resources. Their economic development strategies emphasize the efficient utilization of limited resources, resulting in distinct internal clustering effects. Furthermore, the countries within these regions have adopted multifaceted approaches to foster the growth of healthcare, education, and their economies. Importantly, they emphasize environmental preservation as an integral part of their development efforts. As a result, these regions exhibit relatively higher levels of human development. The climate is conducive in North and South America, and some countries possess relatively robust economies. Not only do they prioritize investments in education, but they also boast world-class healthcare systems, high levels of social welfare, and well-developed public infrastructure. Their commitment to environmental protection alleviates the pressure humanity places on the planet, enhancing human development. Oceania and Asia have made significant strides in development, with notably high growth rates that have begun to reverse the circumstances of underdevelopment. However, the development process in these regions is not without challenges, as resource depletion and ecological degradation exert considerable strain on the planet.

Africa has consistently maintained a trajectory of high PHDI growth. This phenomenon is intricately linked to the policies of assistance by the United Nations and other nations towards Africa. However, the spillover effects of this assistance have yielded only marginal convergence, even though there have been notable improvements in the intersection of development. Africa's PHDI continues to linger at the lower end, with its developmental status not fully reversed. Concurrently, Africa's annual growth rates, when compared to other indices, persist at the lowest levels. The genesis of this phenomenon can be attributed to the fact that most African countries are categorized as developing nations, with most of their industries situated in the primary stage of development. Consequently, their economic advancement remains relatively underdeveloped, characterized by a significant reliance on fossil fuels and a relatively severe level of environmental pollution. While notable progress has been achieved in education and healthcare, the developmental stature of these countries lags far behind that of developed nations. So, the region bears a substantial burden of planetary pressures, contributing to its persistently lower development status.

Overall, during the study period, there was some improvement in the development levels across regions. However, the amelioration in developmental status has been juxtaposed with continued environmental pollution concerns. Notably, the year 2019 witnessed an escalation of human-induced pressures on the planet, owing to the impact of the pandemic, resulting in a decline in PHDI across regions. Hence, it is imperative to harness the radiative driving capacity to efficiently mitigate planetary pressures to bolster the elevation of human development levels in these regions.

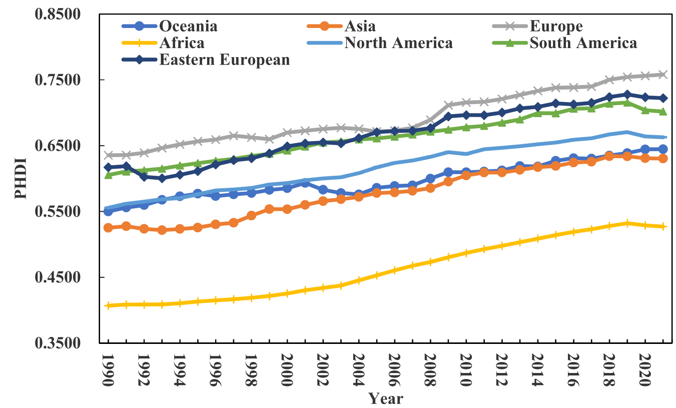

Fig. 2. Basic situation of PHDI in the seven regions.

4.2. Global PHDI spatial disparity analysis

The study spatially decomposes the global PHDI Thiel index into seven major regions, revealing inter-regionally and intra-regionally disparities (see Fig. 3). The PHDI Thiel index ranged between 0.560 and 0.595. A declining trend in global PHDI disparities is observed; however, during 2008–2010, 2016–2018, and 2019–2021, there is a slight upward trend in the PHDI Thiel index. The genesis of these trends may be attributed to the 2008 global financial crisis, causing economic downturns in certain countries. As the economy is a pivotal factor in PHDI calculations, widening economic disparities contribute to increased human development disparities. In 2016, a renewed global economic crisis resulted in disturbances in some countries and regions, such as Brexit, refugee crises, and international events related to climate change, leading to further widening human development disparities. In 2019, the impact of the COVID-19 pandemic threatened human health. Discrepancies in preventive measures among countries and economic and educational stagnation in some nations contributed to a modest increase in PHDI disparities.

From the perspective of differential contributions, the intra-regional disparities in PHDI among the seven global regions are markedly higher than inter-regional disparities. From 1990 to 2021, the contribution rate was consistently maintained within 75%–80%. In contrast, inter-regional disparity contribution values remained between 20% and 25%, exhibiting relatively minor fluctuations. The genesis of these patterns may be attributed to the divergent cultural, economic, and social systems among countries within each region, resulting in variations in intra-regional PHDI. With the ongoing progression of globalization, countries engage in continuous exchanges and collaborations in economic and cultural domains, leading to a gradual reduction in intra-regional disparities. However, disparities in PHDI development levels persist both within and between regions. In summary, intra-regional disparities predominantly determine the direction of global PHDI disparities. Therefore, mitigating intra-regional disparities is pivotal to alleviating spatial imbalances in regional PHDI.

Based on the decomposition of the Thiel coefficient, an analysis of the evolving trends in intra-regional PHDI disparities across the seven major global regions reveals an overall diminishing pattern (see Fig. 4). This trend is attributed to countries' increasing focus on human development, prompting a year-over-year reduction in intra-regional disparities. Concerning intra-regional disparities, the PHDI development gaps among countries within the Asian region are markedly higher compared to other regions. This phenomenon is rooted in Asia being one of the most diverse and populous regions globally, encompassing highly developed economies and relatively impoverished nations. The demographic, economic, social, and cultural diversity may contribute to the imbalance in PHDI. Conversely, European, South American, and Eastern European regions exhibit smaller and relatively closer intra-regional PHDI disparities. Oceania displays minor intra-regional PHDI development gaps attributed to higher welfare levels, minimal disparities in healthcare, education, and income, and intense awareness of environmental sustainability and social equity, resulting in minor intra-regional differences.

Regarding the rate of decline, intra-regional PHDI disparities in Asia have decreased most rapidly. From 1990 to 2021, the Thiel coefficient in the Asian region dropped by 0.129. This decline is attributed to ongoing efforts in Asian countries to promote regional balance in human development, emphasizing infrastructure development and the gradual implementation of a people-centric approach. Despite significant achievements in reducing intra-regional disparities, Asia still faces relatively high disparities, necessitating continued measures to foster regional equilibrium. In contrast, Africa and South America exhibit the most minor variations in intra-regional disparities. Many countries in these regions are developing nations with similar levels of human development. It results in minimal inter-country disparities and comparable planetary pressures, leading to a relatively lower degree of intra-regional human development imbalance.

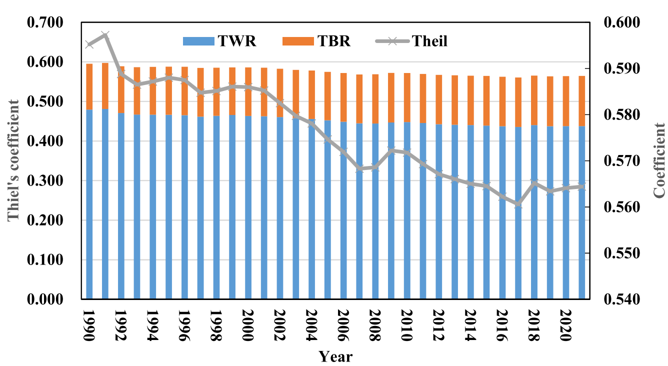

Fig. 3. The case of the PHDI's Thiel coefficient
Note: TWR represents variation within PHDI regions, TBR represents variation between PHDI regions.

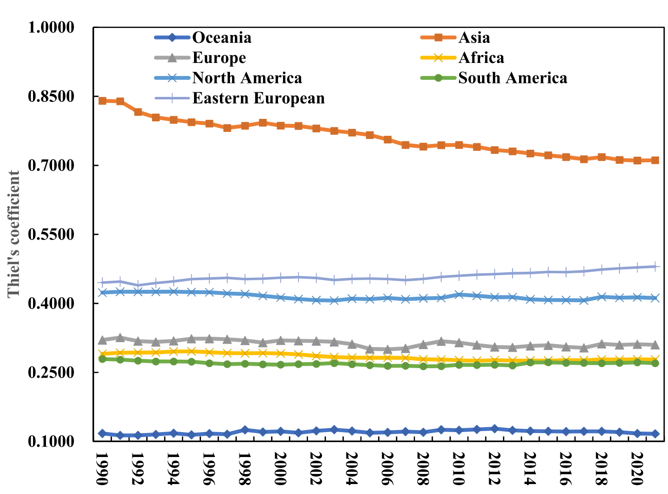

Fig. 4. Intra-regional variation in PHDI.

4.3. Global PHDI driver analysis

4.3.1. Mechanisms of factor decomposition in driving forces

In the face of substantial planetary pressures, identifying the driving factors behind changes in human development levels and discerning the driving effects on global PHDI variations are crucial for unveiling the inherent patterns within the PHDI. Current research on PHDI decomposition is lacking, with existing studies primarily confined to exploring HDI and typically grounded in the Kaya identity. These studies predominantly focus on economic, social, ecological, and technological effects. In conventional HDI research, economic effects primarily gauge economic growth, ecological efficiency effects reflect the carbon emissions intensity associated with economic growth, social effects quantify the contribution of economic growth to human development levels, and technological effects gauge technological proficiency by relating economic output to the emissions generated per unit of economic activity. Nevertheless, a core tenet of the PHDI is its human-centric approach. A significant portion of economic output relies on investments in human capital, and demographic changes also influence economic productivity. Disregarding demographic factors implies overlooking the essential societal characteristics inherent in human development. Notably, existing research has yet to account for the impact of factors such as population size on the PHDI. Clarifying the objectives of human development involves understanding the guidance provided by demographic elements.

Additionally, because economic effects and PHDI are absolute factors, any influence of economic development on the PHDI implies that changes in the ratio of the PHDI to economic effects will inevitably result in alterations in the PHDI when the PHDI remains constant. In summary, the selection of factors after index decomposition remains somewhat fixed yet incomplete, posing a significant challenge to the study of PHDI and hindering its research progress. Consequently, PHDI decomposition should strive for a more comprehensive and objective assessment. Furthermore, the decomposition of driving factors should be oriented toward incremental aspects rather than stock elements. To address the shortcomings in factor selection, this paper, in addition to introducing economic and social effects, examines the influence, direction, and degree of factors such as population size and per capita value-added effects on PHDI. In doing so, it provides essential decision-making foundations for effectively formulating policies tailored to human development.

4.3.2. Analysis of drivers based on the GDIM method

In this study, based on the characteristics of the GDIM method, which can measure both absolute factors and potential driving factors, we used the R language to decompose the driving factors of PHDI to obtain economic effects, environmental pressure effects, population-scale effects, social effects, technological effects, per capita welfare effects, per capita added value effects, output carbon emissions effects, and overall effects. This comprehensive approach complements the selection of driving factors after index decomposition, addressing existing indices' limitations that can only decompose individual absolute quantity driving factors. For analytical purposes, we examined the period from 1990 to 2021 and divided it into three sub-periods: 1990–2000, 2001–2011, and 2012–2021 (see Figs. 5 and 6). The factor decomposition results were calculated according to Eqs. (20)–(26).

From 1990 to 2021, economic impact, technological impact, and per capita welfare impact have consistently exhibited positive driving effects on the PHDI. Conversely, environmental pressure impact, social impact, per capita added value impact, and output carbon intensity impact have consistently demonstrated adverse driving effects. The population size effect on PHDI shows a fluctuating trend.

4.3.2.1. Positive drivers of PHDI. Economic effects. The economic effects exhibit a continuous upward trend, increasing from 0.067 in 1990–2000 to 0.269 in 2001–2011, then declining to 0.051 in 2012–2021. Its driving effect was most substantial during 2001–2011, indicating that global economic development enhances human development. The economic impact encourages the creation of more economic output, and the expansion of the output scale leads to an increase in PHDI. It signifies the presence of a noticeable "scale effect" during this period. Consequently, economic recession would inevitably constrain human development.

Technological effects. Technological effects constitute a significant driver of PHDI variation, exhibiting a noteworthy trajectory of change from 0.060 in 1990–2000 to 0.158 in 2001–2011, followed by a decrease to 0.045 in 2012–2021. This pattern depicts an inverted U-shaped trend, suggesting an initial enhancement followed by subsequent attenuation and maintaining an overall positive driving influence on PHDI dynamics. In other words, the rapid advancement of technology typically yields significant growth effects in its early stages; however, as technology matures and becomes more widespread, these growth effects may attenuate. The underlying causes of this phenomenon can be attributed, on the one hand, to the time human society requires to adapt to new technology and fully harness its potential. In the initial phases of technological adoption, individuals may necessitate time to assimilate the new technology, adjust workflow processes, and acquire relevant education and training, thereby driving a continuous increase in the PHDI. On the other hand, as technology matures and its adoption proliferates, the growth rate in output and benefits may decelerate. Concurrently, the societal adaptations and the educational effects might diminish. Furthermore, technological development often exhibits volatility and cyclical patterns, which could augment technological effects for a period, followed by attenuation as technological cycles fluctuate, resulting in a reduced impetus for advancement. Additionally, technological progress fosters productivity improvements, which, in turn, propels human development. However, due to varying degrees of emphasis on technological innovation in different countries, stemming from diverse demands associated with traditional economic development methods, global disparities in PHDI development persist. Therefore, actively fostering technological innovation and engaging in research and development endeavors to achieve technological advancements facilitates the transformation of global economic development patterns and constitutes a practical pathway to elevate the overall human development levels globally (Wang and Jiang, 2020).

Per capita welfare effects. The per capita welfare effect on PHDI demonstrates a positive driving force, with the driving impact progressively increasing over the years. Specifically, it rose from 0.016 in 1990–2000 to 0.023 in 2001–2011, further escalating to 0.037 in 2011–2021. The genesis of this influence may be twofold: firstly, an elevation in per capita welfare implies an improved quality of life, encompassing enhanced education, healthcare, and social welfare. It fosters the identification of more sustainable resource management and utilization practices, alleviating planetary pressures. Secondly, a higher level of welfare correlates with an elevated quality of life and health status (Guzel et al., 2021). A populace in good health possesses a greater capacity to contribute to society and the economy, thereby aiding in enhancing PHDI. Furthermore, improving per capita welfare can lead to increased focus on environmental protection and sustainable development, prompting measures to reduce environmental pollution and resource overexploitation. It, in turn, mitigates planetary pressures, facilitating an elevation in PHDI.

4.3.2.2. Negative drivers of PHDI. Environmental pressure effects. The environmental pressure effect, transitioning from −0.043 in 1990–2000 to −0.119 in 2001–2011, subsequently decreasing to −0.032 in 2012–2021, manifests a general trend of initial augmentation followed by attenuation, overall reflecting a negative influence. It indirectly suggests that environmental pollution induces substantial planetary pressures, impeding human development. The genesis of this effect lies in the global pursuit of achieving higher outputs with lower inputs, often neglecting environmental impacts. Despite some countries gradually recognizing the unprecedented pressure humanity exerts on the planet (Rockström et al., 2009), necessitating a comprehensive reevaluation of the interdependence and interactions between humans and nature, measuring to facilitate environmental degradation, the environmental pressure effect persists. Urgent actions from more nations are imperative to address this ongoing challenge.

Social effects. The social effect exhibits an opposing driving force on PHDI, displaying a trend of initial augmentation followed by attenuation. The contribution of the social effect transitioned from −0.041 in 1990–2000 to −0.109 in 2001–2011, subsequently decreasing to −0.031 in 2012–2021. It indicates a diminishing inhibitory impact of the social effect on PHDI enhancement over time. The genesis of this phenomenon may be attributed to a global trend of industrial development and lower awareness of environmental issues from 1990 to 2000. The overall inadequate social investment during this period led to a lag in human development. Starting in 2010, rapid societal development globally, accompanied by a strong emphasis on economic growth in various countries, resulted in some nations exhibiting a social effect tendency toward heavy industrialization and extensive economic development. Ultimately, this led to a transformation in the global energy structure, increased planetary pressures, and a significant decline in human welfare. Post-2011, the international community implemented environmental protection legislation and policies, such as the Paris Agreement and the United Nations Sustainable Development Summit. These measures may have prompted increased attention from businesses and governments on reducing planetary pressures. Consequently, the inhibitory effect of the social component on PHDI enhancement exhibits a diminishing trend. Therefore, adjusting societal structures and improving the human development environment are crucial means for the destiny of humanity.

Per capita added value effects. The per capita added value effect shifted from −0.010 between 1990 and 2000 to −0.122 in 2001–2011, declining to −0.012 from 2012 to 2021—the seemingly counterintuitive negative driving impact of the per capita added value effect on PHDI warrants explanation. According to Vaninsky (2014), per capita added value is a relative indicator, representing the ratio of economic effect to population scale effect. Variations in these two indicators will affect their respective PHDI changes, likely asynchronous. Furthermore, economic effects are interrelated with other indicators, leading to the allocation of their variations into other indicators. Only self-induced variations are considered when assessing their impact on the PHDI changes, while the remaining portion is attributed to the influence of other variables. Hence, this interconnectivity of per capita added value potentially has a negative influence.

Output carbon intensity effects. The variation in carbon output intensity effect has gradually intensified, transitioning from −0.016 in 1990–2000 to −0.052 in 2001–2011 and decreasing to −0.009 in 2012–2021. This can be attributed to the proactive response of countries worldwide to the United Nations Global Climate Statement in 2010, prompting concerted efforts to alleviate the state of economic development and environmental degradation. Nations undertook responsibility for human development, adjusting their energy structures and formulating emission reduction measures. However, significant planetary pressures ensued due to the burgeoning global energy demand. Moreover, the varying degrees of adjustments and policy formulations across nations in response to technological and energy transitions contributed to the intensification of the carbon output intensity effect. Improving environmental quality is crucial to ensuring sustainable development (Akram et al., 2023). Without reinforced measures, the driving impact of the carbon output intensity effect is anticipated to strengthen further.

4.3.2.3. The other driving factors of PHDI. Population scale effects. The influence of population size on PHDI demonstrates a fluctuating trend. The population size effect transitioned from −0.005 in 1990–2000 to 0.001 in 2001–2011, decreasing to −0.012 in 2012–2021. During 1990–2000, the population size negatively impacted PHDI, signifying that in the early stages of development, the increase in population size, coupled with a mismatch in public infrastructure and living conditions, resulted in a decline in human development levels. The period of 2001–2011 indicated a positive driving force, suggesting that with the proliferation of supporting facilities, population growth stimulated technological and managerial changes, contributing to a more efficient supply and utilization of resources (Repetto, 1989). The demographic dividend was evident, significantly propelling the elevation of human development levels. However, as the population size expands to a certain extent, environmental issues may negatively impact development. Environmental pollution, resource scarcity, and ecosystem collapse could contribute to a decline in PHDI. It underscores the importance of aligning population growth with environmental protection and resource management within the framework of sustainable development. It ensures that human development can persistently and healthily progress over the long term (Hanif and Gago-de-Santos, 2017).

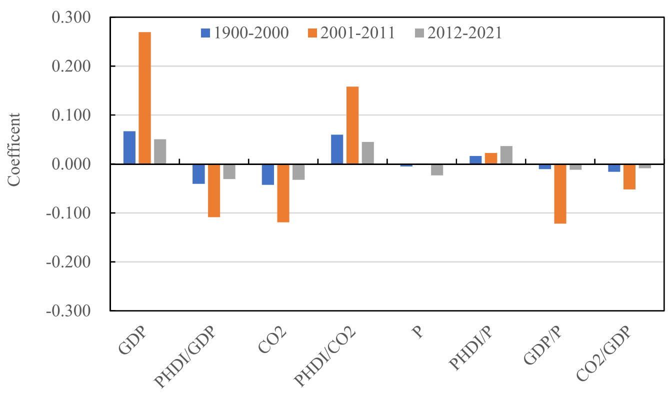

Fig. 5. Phased factor decomposition.

4.3.3. *Dynamic impact analysis of PHDI changes*

In light of the phased analysis conducted, aimed at providing a comprehensive examination of the dynamic influences on the PHDI from 1990 to 2021, this paper adopts 1990 as the base year to calculate the cumulative contribution of various driving factors to PHDI variations. Fig. 7 shows that the PHDI experienced continuous growth from 1990 to 2021. However, post-2009, the rate of PHDI increase gradually decelerated, and in 2019, a declining trend emerged, with PHDI accumulating an increase of 0.112 by 2021. The underlying reasons for this trend can be attributed to the impact of the 2008 financial crisis, which resulted in a stagnation in human development processes primarily driven by national economic development and urbanization. This situation subsequently led to lagged effects in the cumulative contributions of factors such as economic indicators. These effects, in turn, caused a change in human development levels around 2009.

Furthermore, various driving factors underwent directional shifts in 2009, mainly due to the significant global fluctuations in human development triggered by the financial crisis. Diverse economic, social, and environmental issues necessitated urgent attention and varying degrees of policy adjustments by different countries, prompting changes in these effects. In 2019, the outbreak of the COVID-19 pandemic not only increased planetary pressures but also had adverse impacts on global health, economy, education, and social inequality, lowering PHDI. On the one hand, the pandemic strained global healthcare systems, leading to shortages of hospital beds and medical resources, negatively affecting human health and life expectancy (Shrestha et al., 2020), thereby influencing the health dimension of PHDI. On the other hand, to control the pandemic, countries implemented lockdowns and restrictions, resulting in economic downturns and widespread unemployment (Góes and Gallo, 2021), negatively impacting the income dimension of PHDI. Additionally, school closures and educational disruptions during the pandemic may affect students' future potential, thereby influencing the education dimension of PHDI. Simultaneously, the pandemic disproportionately affected vulnerable populations, exacerbating inequalities, also reflected in PHDI.

Therefore, in human development, it is imperative to prioritize economic stability, reduce the pressure exerted on the planet by human activities, and implement stringent policies to ensure the rapid advancement of global human development. In terms of cumulative contribution values, firstly, among the cumulative contribution values of the positively driving factors, the ranking of cumulative contribution

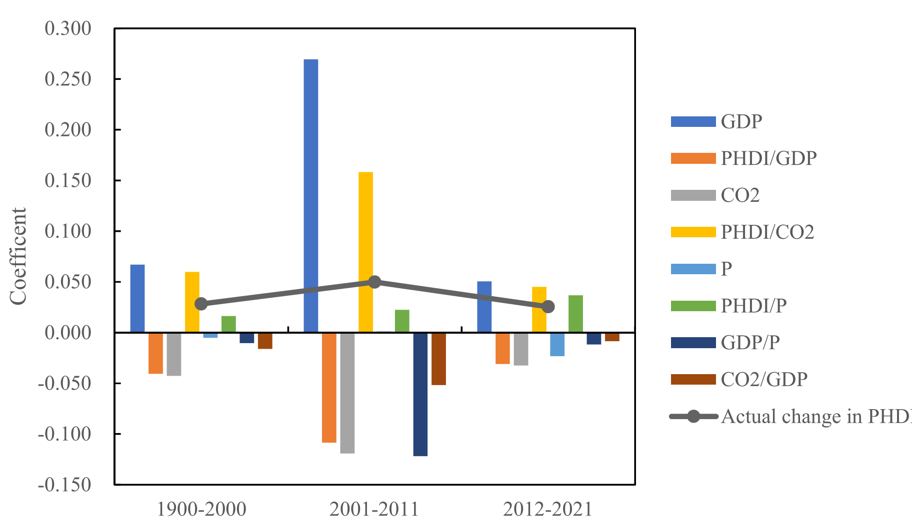

Fig. 6. Decomposition of drivers.

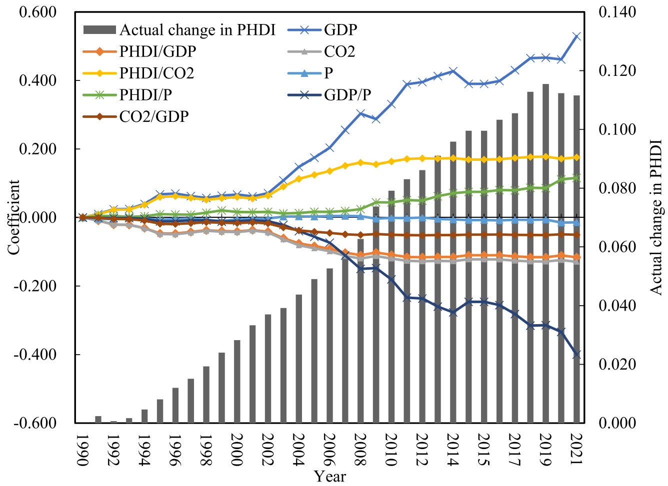

Fig. 7. Cumulative contributions of PHDI from 1990 to 2021.

values is as follows: economic, technological, and per capita welfare effects. As of 2021, economic, technological, and per capita welfare effects have increased PHDI by 0.529, 0.175, and 0.116, respectively. Secondly, among the cumulative contribution values of the negatively driving factors, as of 2021, social effects, environmental pressure effects, per capita growth effects, and the output carbon emissions intensity effects have led to a cumulative decrease in PHDI by 0.116, 0.124, 0.400, and 0.050, respectively. Thirdly, among the other driving factors, the impact of population size effects on PHDI is relatively weak. From 1990 to 2002, it harmed PHDI; from 2003 to 2008, the population size effect rebounded, showing a positive driving effect, and then from 2009 to 2021, its effect on PHDI once again turned negative, with a cumulative contribution value of −0.014 in 2021. The above results indicate that the economic, technological, and per capita welfare effects in countries worldwide have not yet fully promoted PHDI and still have significant room for improvement. Strategies to mitigate environmental pressure effects, social pressure effects, and output carbon emissions effects should be prioritized in line with the global sustainable development agenda. Therefore, based on the results above, in the future, countries worldwide should primarily focus on enhancing technological capabilities, minimizing environmental pollution to maximize economic benefits, and thus mitigating global pressures to elevate human development levels.

4.3.4. Robustness testing

In this study, the GDIM is employed to decompose the PHDI into multiple driving factors, facilitating an analysis of the influences exerted by various absolute and relative determinants on PHDI. However, it is imperative to acknowledge that the accuracy, scientific rigor, and reliability of GDIM's outcomes may be compromised due to the potential presence of multicollinearity among the variables during the decomposition process, posing concerns regarding the validity of the results. Consequently, to ensure the precision of the GDIM outcomes, this section conducts robustness tests. Robustness tests entail exploring the impact of driving factors on PHDI through regression estimates. Nevertheless, owing to the strong interdependencies among the eight explanatory variables, the issue of multicollinearity is highly probable, thus potentially affecting the results. To enhance the accuracy of the estimation outcomes, this study follows the approach You and Yan (2017) proposed in constructing a stepwise regression model.

Furthermore, logarithmic transformations are applied to the variables to mitigate heteroscedasticity concerns, following the methodology described by Shen (2012) to investigate further the effects of driving factors on PHDI. Presented below are the results of the stepwise regression analysis. Presented below are the results of the stepwise regression analysis.

Table 1
Stepwise regression results.

Variables	ln *phdi*
$\ln x_5$	0.390***
	(7.751)
$\ln x_8$	−0.043***
	(-3.495)
_cons	−9.810***
	(-9.739)
r2_a	0.987

Note: *p < 0.1, **p < 0.05, ***p < 0.01.

$$\ln phdi = \alpha + \theta_1 \ln x_1 + \theta_2 \ln x_2 + \theta_3 \ln x_3 + \theta_4 \ln x_4 + \theta_5 \ln x_5 + \theta_6 \ln x_6 + \theta_7 \ln x_7 + \theta_8 \ln x_8 + \varepsilon \tag{27}$$

Among them, $x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8$ denote the economic effects, social effects, environmental pressure effects, technological effects, population size effects, per capita welfare effects, per capita value-added effects, and output carbon effects, respectively. θ_i denotes the coefficient of variables. ε denotes the error of the regression.

The present study, through a comparative analysis of the outcomes derived from the stepwise regression approach and those from the GDIM, has discerned that the population size effect exhibits a positive driving role. Table 1 aligns with the enhanced impact of population size observed in the cumulative effects of GDIM for the years 2020–2021 on the PHDI. Additionally, the output carbon intensity effect exerts a negative driving influence on PHDI, which is fundamentally consistent with the outcomes generated by GDIM. These congruent findings further underscore the robustness of the GDIM results. It is imperative to underscore the significance of precise and reliable analytical methodologies in furnishing critical insights for effective policymaking and decision-making processes, particularly in elevating human development levels.

5. Discussion

In general, empirical research findings can serve as a valuable tool for addressing global challenges, enhancing regional HDI improvements, and advancing global sustainability. This study affirms the conclusions drawn by Wang and Zhai (2018): disparities in human development levels among the seven major global regions remain a significant characteristic of uneven global human development. However, their contribution to overall spatial disparities has diminished. Simultaneously, contributions to regional disparities are increasing, resulting in an overall reduction of HDI disparities within these seven major global regions. Moreover, this paper utilizes the GDIM method to dissect the driving factors behind human development levels. The results indicate that economic and technological factors exert positive influences, whereas social factors have adverse effects, aligning with the findings of Wang and Jiang (2020). This comprehensive demonstration underscores the feasibility of the GDIM method in examining the driving factors of human development levels. Given that the GDIM method considers both absolute and potential factors related to the research subject, the obtained results are more thorough and profound, overcoming the limitations of conventional index decomposition models. It further emphasizes the method's robust applicability. Additionally, the positive impact of economic effects on human development levels is akin to the conclusions drawn by Carvalho et al. (2016), highlighting the affirmative economic effects of system and economic reforms, significantly influencing various aspects of human development with scientific merit.

In the realm of disparities in human development, the root causes of overall disparities in human development levels exhibit variability. According to Wang and Zhai (2018), a paramount determinant of the global disparities in human development levels among the seven major regions from 1990 to 2014 was the discrepancies in human development levels within these regions. Conversely, this study posits that the primary drivers of overall disparities lie in the variances of human development levels within regions. These discrepant findings can, in part, be ascribed to variations in the methodologies employed for gauging human development levels. Wang and Zhai (2018) primarily utilized the HDI, which assesses human development levels based on health, education, and income. In contrast, this study adopts the PHDI as its metric for evaluating human development levels. The PHDI, while encompassing the conventional dimensions of health, education, and income, goes a step further by integrating indicators related to environmental pollution, resource depletion, and other factors. This integration results in calculating a planetary pressure coefficient, offering a more comprehensive assessment of the HDI. This modification aligns the index with the imperative of environmental sustainability, a crucial facet of contemporary human development analysis. Furthermore, there are distinctions in the scope of research between Wang and Zhai (2018) and the present study. While they primarily scrutinized human development levels across 136 countries, this study, relying on extensive existing data, delves into the human development status of 154 countries, thus broadening the global context of the research. Moreover, this paper incorporates the sustainable development framework within the contemporary discourse on human development levels. By adhering to the United Nations Development Programme's measurement methodology and considering the influence of planetary pressures on human development, the findings elucidate the factors underlying overall disparities in human development in harmony with the prevailing global paradigms of human development analysis.

Regarding innovation, existing literature predominantly employs the HDI to study human development levels, overlooking planetary pressure factors such as environmental pollution and resource consumption, thereby failing to analyze human development (Sagar and Najam, 1998; Bravo, 2014) precisely. This paper utilizes the latest assessment methodology from the United Nations Development Programme to incorporate environmental pollution and resource consumption into the HDI, thereby facilitating a more accurate depiction of human development. Moreover, prior research often focuses on individual countries or provinces with limited global perspectives on human development levels. However, this paper employs the Theil index to measure global disparities in human development across seven major regions. It further dissects the Theil index into intra-regional and inter-regional components to gain insights into the human development status within these seven significant regions. Finally, considering the driving factors of PHDI, we construct a multidimensional GDIM decomposition model encompassing multiple absolute and relative factors. This model enables a more comprehensive and precise quantification of the actual contributions of various factors to the evolution of PHDI under the pressures of our planet. It effectively addresses the limitations of existing index decomposition methods, which exhibit formal dependencies among factors. This approach allows for a more comprehensive and in-depth analysis of the impact of global PHDI driving factors.

6. Conclusions and recommendations

6.1. Conclusions

In pursuit of sustainable development, this study is conducted based on the global PHDI measurement, spatial disparities, and the analysis of driving factors. The research findings indicate the following: Firstly, the development levels of PHDI vary across regions. The specific spatial ranking evolution is as follows: Europe and Eastern Europe exhibit higher levels of human development, North and South America have relatively good human development levels, Oceania and Asia show similar levels of human development, and there is significant room for improvement in human development levels in Africa. Secondly, countries engage in ongoing economic and cultural exchanges and collaborations as globalization advances, gradually reducing overall PHDI disparities. However, disparities in PHDI levels persist between regions and within regions, with intra-regional disparities notably higher than inter-regional ones. Thirdly, concerning intra-regional disparities, the variation in PHDI development levels among Asian countries is significantly higher than in other regions. Europe, South America, Eastern Europe, and Oceania exhibit smaller and relatively close gaps in PHDI development levels. At the same time, disparities within the African and South American regions show the slightest degree of change. Fourthly, economic, technological, and per capita welfare effects consistently exert positive driving effects on PHDI. Meanwhile, environmental pressure effects, social effects, per capita value-added effects, and output carbon intensity effects consistently exert adverse driving effects. The impact of population size effect on PHDI displays a fluctuating trend. Furthermore, in terms of cumulative contribution values, the top three contributors to driving effects are economic, technological, and per capita welfare effects.

6.2. Recommendations

In the face of immense global pressures to enhance human development levels across the seven major global regions and achieve economic Pareto optimality, this paper, rooted in the framework of circular economic development and the "people-centered" sustainable development goals, proposes the following recommendations.

6.2.1. Elevating human development levels and promoting coordinated PHDI development across seven significant regions

For elevating global human development levels, proactive efforts should be made to promote coordinated PHDI development across the seven major global regions. Empirical findings underscore the disparities in PHDI development across regions, with an overall suboptimal level and a tendency of declining development in developed regions transitioning towards developing ones. To address this, when formulating comprehensive development strategies, the emphasis should be on the concurrent progress in economics, education, and healthcare, focusing on embracing the circular economy as a solution to environmental challenges (Cifuentes-Faura, 2022a). Specific measures to enhance human development and promote sustainability include: In the economic sphere, nations should stabilize economic growth and encourage win-win cooperation among regions to advance sustainable and coordinated development strategies. In healthcare and education, developed nations can share their expertise and knowledge to support the development of developing regions. It may involve technology transfer, talent mobility, and knowledge sharing to improve healthcare and education standards. Regarding environmental sustainability, measures to reduce environmental pollution should be tailored to regional differences, economic foundations, and development orientations. Drawing from the European Commission's successful experiences, the establishment of working groups and long-term sustainable development plans can be pursued. These plans can incorporate reward mechanisms for nations that achieve their targets and penalties for those that fail to comply (Cifuentes-Faura, 2022a). Furthermore, promoting green technology innovation and bolstering the utilization of renewable energy resources can be instrumental in fostering sustainability (Wang et al., 2023). These efforts can contribute to reducing global pressures and subsequently increase PHDI levels.

6.2.2. Eliminating disparities in human development regions and promoting spatial equilibrium through policy orientation

To achieve regional development balance while adhering to established development goals, nations worldwide should seek equilibrium within their development trajectories. It entails bolstering economic and social exchanges, increasing investments in healthcare and education, actively enhancing infrastructure, and initiating structural adjustments in regions heavily reliant on heavy industry for economic growth to mitigate the impact of planetary pressures on human development. In collaborative efforts, two key aspects should be considered:

Firstly, by addressing regional disparities, well-developed nations can support the development of countries in less favorable circumstances through one-on-one assistance. This support might take the form of tailored industrial, technological, human resource, and resource assistance, focusing on considering the unique characteristics of African regions. This approach can help narrow regional disparities. Secondly, the United Nations 2030 Agenda for Sustainable Development, aimed at establishing public health systems, eradicating poverty, promoting decent work, and steering industrial, transportation, and consumption systems toward sustainability, has emerged as a highly applicable tool (Cifuentes-Faura, 2022b). Therefore, progress toward sustainable development and the reduction of disparities in PHDI, both within regions and between them, can only be realized by establishing and refining relevant legal frameworks, clarifying government authorities and responsibilities, aligning the interests of governments across nations, balancing short-term and long-term interests, and enhancing the stability of cross-regional cooperation and governance mechanisms. Specifically, lessons can be drawn from regions with high levels of human development, where corresponding policy support measures can be implemented. For instance, the nations could impose carbon emission taxes on regions with severe pollution, urging them to optimize their industrial structures and alter their energy consumption patterns. Furthermore, tailored national policies should be introduced to address intra-regional differences. For regions with significant disparities like North America and Asia, promoting the "Internet+" approach and fostering competent healthcare, education, and environmental systems can encourage economic and cultural exchanges within the region, thereby propelling human development. In regions with smaller intra-regional disparities, such as Europe, emphasis can be placed on developing intelligent management systems to encourage countries to improve their development structures, elevate human development levels, and subsequently expand the radiating effects of sustainable development to promote spatial equilibrium.

6.2.3. Comprehensively considering PHDI driving factors to enhance global human development through a multifaceted approach

In addressing the challenges faced by global human development and charting a rational and viable development path that promotes harmonious coexistence between humans and nature while advancing sustainable development, it is imperative to deepen global structural reforms and prioritize the concept of harmonious coexistence between humans and nature. This approach should also balance industrial development, human development enhancement, and alleviating planetary pressures. Given the varying degrees and directions of influence exhibited by different factors, several key strategies can be adopted:

Economically, nations worldwide should actively foster economic development that maximizes economic output with minimal environmental pollution. While pursuing economic growth, efforts should also be made to guard against economic crises. Regarding population, well-developed human capital and the shift towards clean energy are instrumental in promoting a cleaner environment (Uche et al., 2023). Nations should proactively implement talent acquisition programs, improve living conditions, and offer competitive salaries to incentivize talent retention, thereby driving technological research and development and alleviating regional planetary pressures.

In the realm of technology, international cooperation and knowledge exchange are crucial. Developed nations should prioritize the development of critical industries such as next-generation information technology, renewable energy, and advanced materials while expanding energy efficiency and environmental protection efforts. It includes improving energy consumption patterns and reducing carbon emissions per output unit, mitigating regional planetary pressures. Conversely, less developed nations should actively introduce advanced technologies to phase out outdated capacity, promote the development of new industries, and nurture strategic emerging industries to elevate human development levels. Environmental considerations should be at the forefront of government agendas. Governments should establish robust environmental protection regulatory systems, strengthen energy efficiency and law enforcement, ensure effective energy conservation and environmental protection laws, regulations, rules, and mandatory standards, and rigorously address illegal energy use and environmental violations. Simultaneously, building upon existing environmental policies, continuous innovation, and heightened environmental awareness campaigns should be pursued to facilitate improvements in human development levels. Maintaining the current level of welfare and continually enhancing the quality of life in each country through ongoing innovation is essential. Additionally, societal structures should be adjusted to prioritize harmonious coexistence between humans and nature. The concept of sustainable development should be integrated into the development strategies of all nations, thereby mitigating the conflicts between societal resource demands and human development.

6.3. Limitations and future research directions

In this study, certain limitations are evident. On the one hand, there remains a need for further exploration into the driving factors of the PHDI across various regions. On the other hand, research at the national level demands immediate supplementation. Nevertheless, the methodology employed in this study presents novel avenues for future analysis using the GDIM. The issues above can be addressed by examining specific individual regions or countries.

Furthermore, prospective research could consider the following dimensions to mitigate the constraints of this study. Firstly, future investigations may employ alternative methodologies to investigate regional disparities in PHDI. Secondly, attention could be directed towards the national level, such as measuring PHDI across various provinces in China, thereby elucidating intra-regional disparities in human development and exploring regional variations in human development levels. Through regional studies, effective pathways for mitigating planetary stress and enhancing human development can be unearthed. Thirdly, a focus on sustainable development is warranted in forthcoming research, delving deeper into methods for elevating human development levels.

CRediT authorship contribution statement

Xinyan Lian: Formal analysis, Writing – original draft, Writing – review & editing. **Jingquan Chen:** Conceptualization, Writing – review & editing. **Zongping Fu:** Writing – review & editing, Funding acquisition

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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